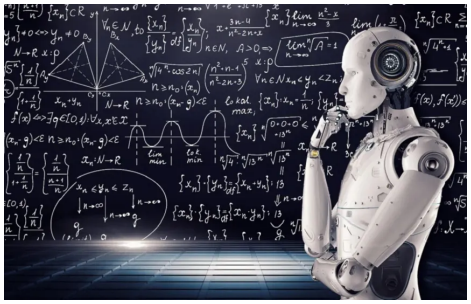
# What is Artificial Intelligence (AI)?

# Purpose of the Lesson

After this lesson, you will understand the fundamental definition of Artificial Intelligence, the core elements that make AI 'intelligent', and how AI differs from simple automation.

# 1. The Definition of AI

Artificial Intelligence (AI) is a field of computer science that focuses on creating systems or machines that can perform tasks that, if done by humans, would require intelligence. It can be seen as the science and engineering of creating intelligent machines, specifically intelligent computer programs.



In short, AI is about making computers so 'smart' that they are able to perceive their environment, reason about information, learn from experiences, and make decisions or take actions.

Not all AI systems perform all of these functions at the same time. Some only perceive and reason, while others may also learn or act.

Analogy for beginners: Think about how to teach a dog to play fetch. You throw the ball and encourage the dog to bring it back. After a few attempts, the dog learns what you want. AI systems learn from examples and experiences in a similar way, but instead of rewards, they use data and algorithms.

# 2. The Core Elements of AI

To understand the definition, we look at the most important characteristics:

### A. Simulation of Intelligence

AI aims to replicate or simulate aspects of human intelligence. This includes abilities such as:

- Recognizing patterns.
- Troubleshooting.
- Understanding language.
- Learning from experience.
- Making decisions or recommendations.

AI simulates these abilities through mathematical models and algorithms, not through human understanding or consciousness.

### B. Task performance

The tasks that AI systems perform range from simple (such as sorting emails) to very complex (such as diagnosing illnesses or driving). What makes these tasks 'intelligent' is that they traditionally require human expertise or brainpower.

AI systems are usually designed for specific tasks rather than general human-like intelligence.

### C. Autonomy and adaptation

A core component of AI is its ability to operate with a certain degree of autonomy. This means that the system can make choices or take actions without being explicitly programmed for every possible scenario. Additionally, AI systems can adapt, improving their performance as they gain more experience or process more data.

The level of autonomy varies widely, from fully human-controlled systems to systems that act independently within defined boundaries.

## 3. How a System Becomes 'Intelligent'

A system is considered to have artificial intelligence if it can perform the following four fundamental steps:

- Perceive: Collect information from the environment, for example via sensors, microphones or cameras.
- Reasoning: Processing and interpreting that information to draw conclusions or make decisions.
- Learn: Improve performance over time by learning from data or experience.
- Act: Taking actions or making recommendations based on reasoning.

Not every AI system includes all four steps; many systems focus on only a subset.

## 4. The Role of Data and Algorithms

AI relies on two main components:

- Data: Information from the world that is used for learning.
- Algorithms: Sets of instructions or rules that are used to process that data, recognize patterns, and make predictions or decisions.

AI uses mathematical formulas and algorithms that use statistics and probability to mimic cognitive functions such as decision-making.

AI is an umbrella term that includes fields such as Machine Learning (ML) and Deep Learning (DL):

- Machine Learning: Enables machines to learn from data and improve their performance, leading to data analysis and optimization. This learning process often consists of a training

phase (where the model learns from existing data) and an evaluation phase (where the model has to classify new data to achieve the desired result).

- Deep Learning: This is inspired by the functioning of the human brain, biological neural structures and implemented using artificial neural networks. These networks are powerful at processing and interpreting vast amounts of complex data, which is necessary for tasks such as classification, prediction, and image processing.

## 5. What AI is NOT (Key Misconceptions)

To avoid confusion, it's important to know what AI is not:

- Not Just Automation: Simple automation (like a washing machine that runs through a fixed cycle) is not AI. AI requires a certain amount of 'thinking', where the system analyses, learns or adapts.
- Non-Human Intelligence: AI processes information and makes decisions based on rules and data. AI does not 'feel' or 'think' like a human, it lacks emotions and intuition, consciousness, and self-awareness.

AI intelligence is functional, not experiential.

## 6. AI in Everyday Life (Examples)

You probably interact with AI on a daily basis without realizing it:

| AI Application | What it does (Requires Intelligence) | Core capabilities |
|---|---|---|
| Voice assistant (Siri/Google) | Recognizes spoken language, understands the intention, looks for an answer and responds. | Perceiving (speech), Reasoning (understanding), Acting (answering). |
| Spamfilters | Scans patterns in emails and decides which ones are unwanted, learning from millions of examples. | Reasoning, learning. |
| Netflix Recommendations | Analyzes viewing behavior and predicts what you will like most. | Learning, Reasoning (prediction). |

| | | |
|---|---|---|
| Face unlock | Compares your face to saved footage to grant access. | Perceiving (image processing), Reasoning (comparison). |
| Self-driving cars | 'See' the way, read signs, and avoid obstacles. | Perceiving (sensors), reasoning, acting. |
| Financial Fraud Detection | Monitors and detects suspicious credit card payments in real-time. | Learning, Reasoning (detection). |
| Chess computers | Analyze the board and choose the optimal move, often faster than a human. | Reasoning. |

Imagine an AI system as an extremely organized, high-speed librarian. That librarian has read billions of books (data) and, thanks to complex indexes (algorithms), can quickly find the right information, discover patterns in what people read, and autonomously decide which new book to recommend to you based on your history, but the librarian doesn't 'feel' joy in finding the perfect book.

This illustrates both the power and the limits of Artificial Intelligence.

Artificial Intelligence is about data-driven systems that can perceive, reason, learn, and act. It is not human intelligence, but it enables machines to solve complex problems at scale.